Advanced Dialogflow Chatbot: Loan + Investment Assistant

AI-Based Financial Guidance using Dialogflow ES

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PROJECT INTRODUCTION

Bridging the Financial Knowledge Gap

The Problem

Most users struggle to understand loan eligibility criteria, EMI calculations, and investment planning — leading to poor financial decisions.

Our Solution

An AI-powered finance chatbot using Dialogflow ES that delivers instant, conversational guidance on loans, EMIs, and investments (SIP).

Core Objectives

- Automate finance-related responses using NLP
- Improve user interaction through conversational AI
- Demonstrate practical chatbot and intent-based technologies

AI & NLP in Finance

What is AI in Finance?

Artificial Intelligence (AI) refers to the use of smart algorithms and machine learning models to automate financial decision-making, analyze large datasets, and improve efficiency in banking and investment systems.

What is NLP (Natural Language Processing)?

Natural Language Processing (NLP) is a branch of AI that enables machines to understand, interpret, and respond to human language in a conversational manner.

Role of AI & NLP in Finance

- Automates customer interactions through chatbots
- Provides real-time financial advice
- Enhances fraud detection and risk analysis
- Improves decision-making using data-driven insights
- Enables personalized investment recommendations

Applications in Your Project

- Chatbot understands user queries like: "Can I get a loan?" and "Suggest low-risk investment"
- NLP processes user intent using Dialogflow
- AI provides accurate responses based on financial logic
- Reduces human effort and increases response speed

Real-World Examples

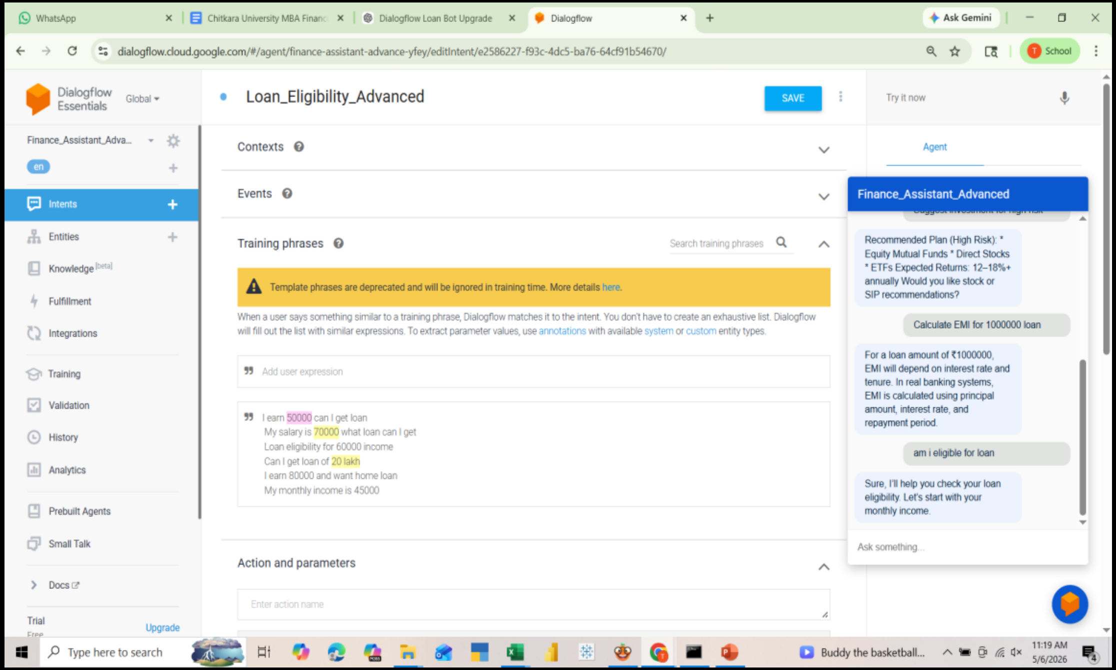
- Banking chatbots for customer support
- AI-based loan approval systems
- Robo-advisors for investment planning
- Voice assistants for financial queries

Why It Matters

Faster	Smarter
More accessible	User-friendly

MODULE 1

Loan Eligibility Intent



Intent: `Loan_Eligibility_Advanced`

Assesses loan eligibility based on user-provided monthly income and financial profile.

Sample Training Phrases

- "I earn 50000 can I get a loan?"
- "Loan eligibility for 60000 income"
- "I earn 80000 and want a home loan"

Parameters

- `income` → `@sys.number` (Required)
- `loan_amount` → `@sys.number` (Optional)

Eligibility Factors

Monthly income · Credit score · Existing EMIs · Employment type · Repayment history

Age Eligibility

MODULE 2

EMI Intent

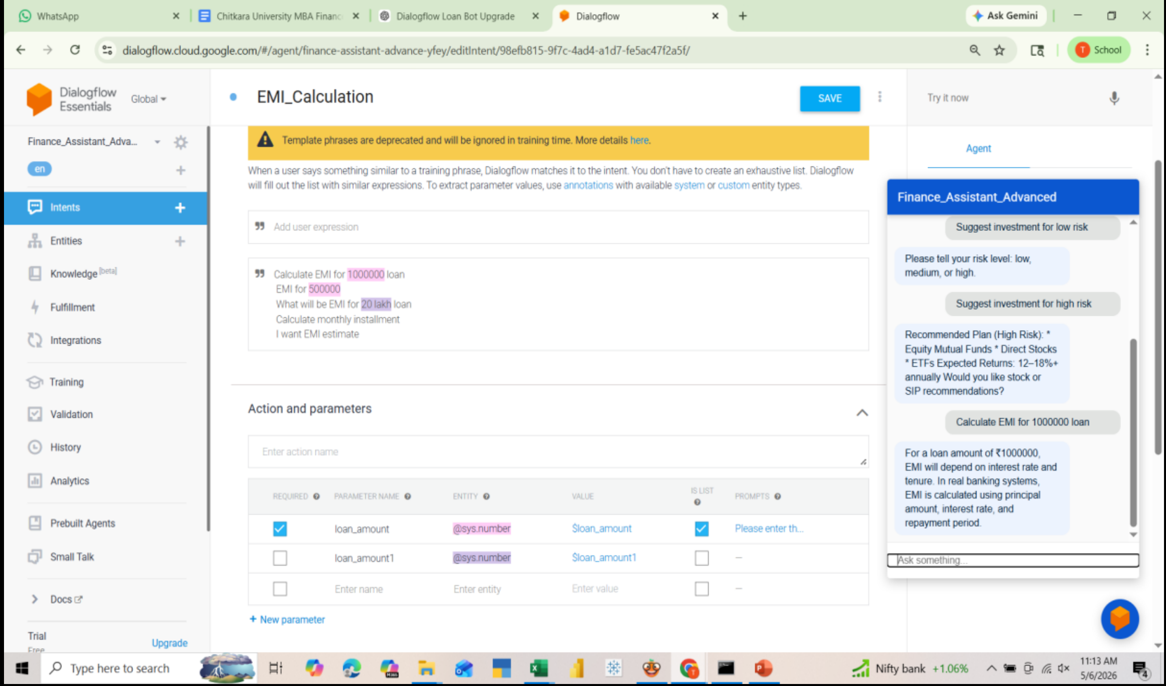
Intent: `EMI_Calculation`

Helps users understand possible EMI ranges and loan affordability based on their requirements.

Bot Calculates EMI
Use this EMI Formula : $EMI = \frac{P \times R \times (1+R)^N}{(1+R)^N - 1}$

Parameter

`loan_amount` → `@sys.number`



MODULE 3

Investment Intent

Intent: `Investment _Advanced` **I Custom Entity:** `risk_level`

Conservative Preference

Fixed Deposits, Bonds, Debt Mutual Funds

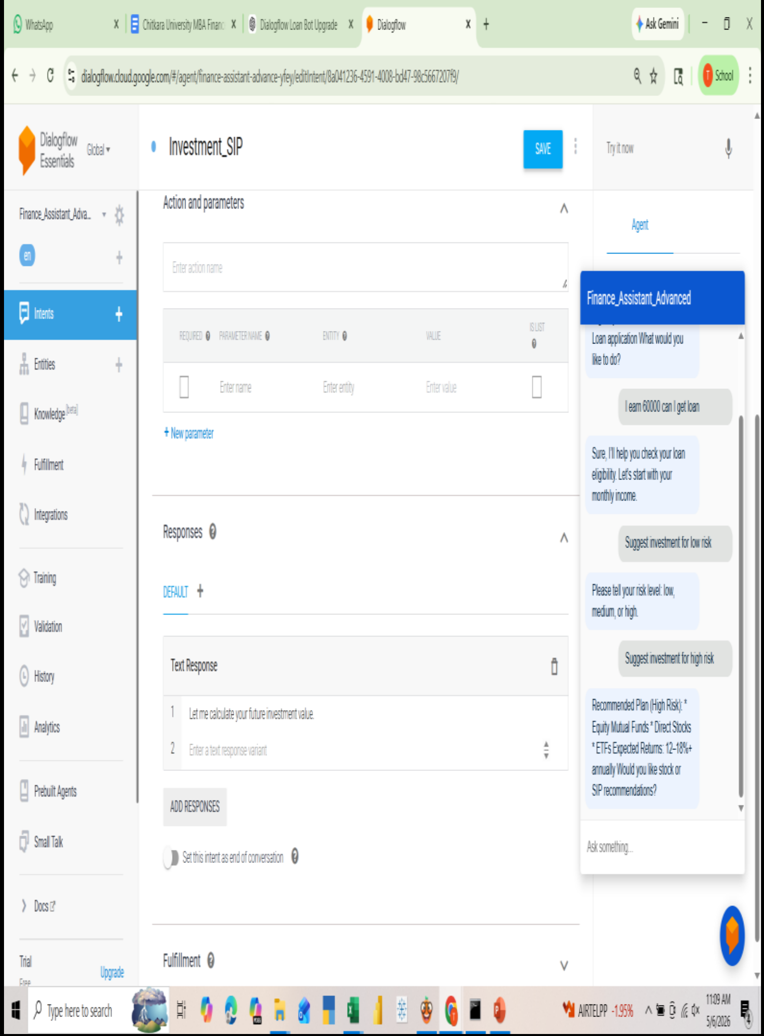
Balanced Preference

Balanced Funds, Index Funds, Hybrid Schemes

Growth Preference

Stocks, Equity Mutual Funds, Derivatives, SIP (Systematic Investment Plan)

Sample Phrases: "I want safe investment" · "I can take high risk" · "Investment for medium risk"



LOGIC FOR EMI CALCULATION

Bot Calculates EMI

Use this EMI Formula : EMI = \frac{P \times R \times (1+R)^N}{(1+R)^N - 1}

Where:

- P = Loan Amount
- * R = Monthly Interest Rate
- * N = Number of Monthly Instalments
- Example Calculation
- If:* Loan = ₹5,00,000
- * Interest = 10%
- * Tenure = 5 years
- Then:
- * Monthly EMI ≈ ₹10,624
- * Total Payment ≈ ₹6.37 Lakhs

LOGIC FOR SIP CALCULATION

Bot Calculates SIP Return

SIP Formula:

$$M = P \times \frac{\left(1+\frac{r}{n}\right)^{nt}-1}{\frac{r}{n}} \times \left(1+\frac{r}{n}\right)$$

Where:

P = Monthly Investment
* r = Annual Return
* t = Time in years
* n = 12 months

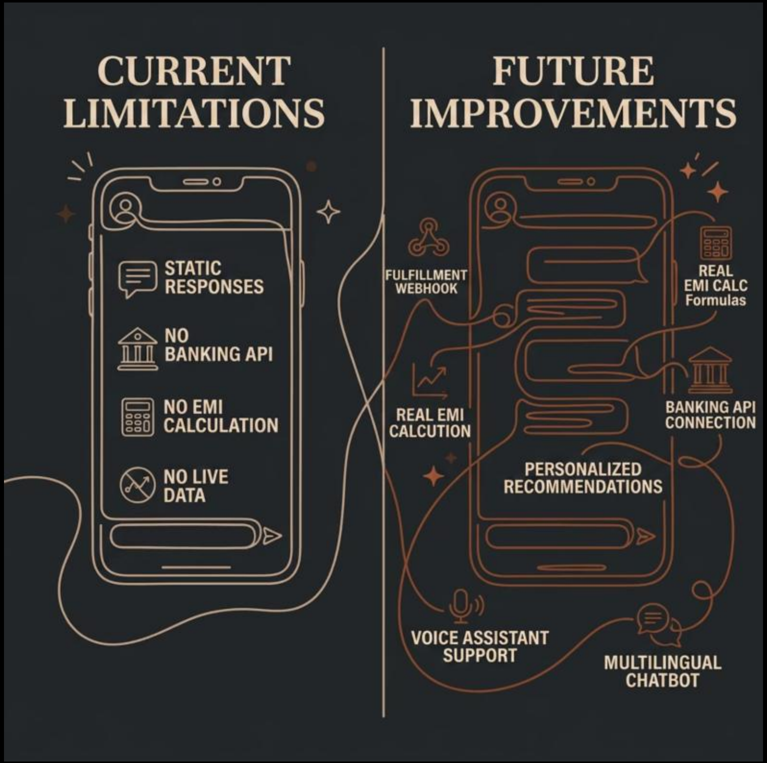
Example Output

If:* SIP = ₹5,000/month
* Duration = 10 years
* Return = 12%

Then:
* Total Invested = ₹6 Lakhs
* Estimated Returns ≈ ₹5.6 Lakhs
* Final Amount ≈ ₹11.6 Lakhs

ROADMAP

Limitations & Future Improvements



What We're Building Towards

The current version demonstrates core NLP capabilities. Future iterations will connect to live financial APIs, enable real-time calculations via webhooks, and support voice and multilingual interactions for broader accessibility.

Next milestone: Integrating a Fulfillment Webhook to enable dynamic, real-time financial computations.

Conclusion

Key Achievements

1. Built an intelligent chatbot using Dialogflow ES
2. Applied NLP to understand user queries accurately
3. Automated financial guidance with real-time responses
4. Improved user experience through simple conversation flow

Business Impact

- Reduces dependency on human financial advisors
- Saves time and operational cost for financial institutions
- Provides instant, 24/7 financial assistance
- Makes financial knowledge accessible to a wider audience

Learning Outcomes

- Practical understanding of AI and NLP in finance
- Hands-on experience with chatbot development
- Improved knowledge of financial decision-making concepts
- Integration of technology with real-world applications

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- https://github.com/nakshatra2654mba25fin-sketch/ai-finance-expense-automation-system/blob/main/Lab_1_2_Bias_Detection_Loan_Approval.ipynb